

ADARSH PATEL

AI Engineer | Generative AI | Full Stack Developer

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PROFESSIONAL SUMMARY

AI Engineer with 4+ years of experience designing and shipping production-grade Generative AI platforms, including LangGraph multi-agent systems, RAG pipelines, NLQ-to-SQL engines and real-time voice AI on AWS. Team Lead on an enterprise agent platform with a No-Code Agent Builder, WhatsApp/Slack integrations, RBAC and SSO, which reduced reporting time by 75% and document search time by 80%. Strong full-stack (FastAPI, Spring, React) and production ML (LightGBM forecasting) background, owning the full lifecycle from data and model development to deployment, monitoring and scaling. AWS Certified Solutions Architect – Associate.

TECHNICAL SKILLS

AI	LangChain • LangGraph • RAG Pipelines • Multi-Agent Orchestration • OpenAI • Gemini • Hugging Face • TensorFlow • PyTorch • Pinecone • NLQ-to-SQL • MCP Tool Integration
ML	Feature Engineering • Supervised Learning • SQL Time-Series Forecasting • LightGBM • XGBoost • Scikit-learn • Pandas • NumPy • Model Evaluation (WMAPE, MAE, RMSE)
Languages	Python • Java • JavaScript • SQL
Backend	FastAPI • Flask • Spring Boot • Node.js (Express) • REST APIs • WebSockets • JWT • SSO • RBAC • Microservices
Frontend	React.js • Tailwind CSS • AG-Grid • AG-Charts • Module Federation • Webpack
Cloud / DevOps	AWS ECS/Fargate • EC2 • S3 • RDS • Lambda • CloudFront • ALB • Route 53 • Docker • GitHub Actions • CodePipeline • CodeBuild • Nginx
Databases	PostgreSQL • MySQL • MongoDB • Redis • Snowflake • AWS Redshift • SQLAlchemy • Hibernate

PROFESSIONAL EXPERIENCE

AI Engineer

BITTWOBYTE Private Limited • Rewa, India

May 2023 – Present

AI/ML Engineering

- Architected an enterprise Agentic AI platform using LangGraph multi-agent orchestration across analytics, document intelligence, video intelligence and enterprise integrations
- Built a No-Code AI Agent Builder and a voice/chat AI Copilot (Gemini Live API, WebSocket streaming) so users can create workflows and control the platform by speech or text.
- Developed an NLQ-to-SQL engine that turns natural language into optimized SQL with real-time visualizations, plus a KPI Wallet to pin insights as live KPI cards.
- Built RAG systems covering document ingestion, chunking, embeddings and semantic search; designed vector memory with Pinecone and Redis, cutting token consumption by 20%.
- Implemented MCP-style tool integrations connecting agents to enterprise APIs, CRMs and databases, and integrated agents with WhatsApp and Slack for remote workflow execution.
- Built AI Tutor (adaptive learning with an AI Student Personality Engine) and a Resume-JD matching engine for ATS-style skill-gap analysis.
- Delivered an agrochemical demand forecasting platform: lag, rolling, seasonal and calendar features; global LightGBM models validated with rolling time-series CV (WMAPE, MAE, RMSE); FastAPI batch and real-time inference with automated retraining on AWS.

Full Stack & Cloud

- Built microservices with FastAPI, Flask, Spring Boot and Express; implemented JWT auth, Microsoft SSO and multi-tenant RBAC with isolated tenant data.
- Cut API latency by 40% through Redis caching, async processing and connection pooling; improved DB performance by 60% with indexing and query tuning; reduced Snowflake workflow runtime by 35%.

- Built React dashboards with AG-Grid/AG-Charts and Module Federation micro-frontends, reducing reporting dependencies by 70%; added WebSocket streaming for AI responses.
- Provisioned AWS infrastructure (ECS/Fargate, ALB auto scaling, CloudFront) with CI/CD on GitHub Actions and CodePipeline; handled production support, improving recovery speed by 50%.

Software Developer Intern

TECHLOZY · Rewa, India

Oct 2022 – Mar 2023

- Developed full-stack applications using React.js, Node.js, and Express, improving system scalability and responsiveness.
- Built REST APIs for authentication and data operations, improving backend efficiency by 25%.
- Integrated MongoDB and PostgreSQL with optimized schemas, improving query performance.
- Deployed cloud applications using AWS EC2, S3, and Docker with CI/CD running on GitHub Actions.

PROJECTS

AI Multi-Agent Automation Platform

Team Lead / AI Engineer

React | Python FastAPI | LangChain | LangGraph | LLM Models | Pinecone | Redis | Docker | AWS ECS | CloudFront | Nginx

- Built a production-grade multi-agent AI platform with No-Code Agent Builder for RAG, NLQ-to-SQL, video intelligence, and multimodal AI workflows.
- Developed an AI Copilot with voice/chat capabilities and integrated agents with WhatsApp and Slack for enterprise workflow automation.
- Implemented RBAC, Microsoft SSO, and Auto-scaling, reducing reporting time by 75% and document search time by 80%.

AI Tutor: Adaptive Education Platform

FullStack / AI Engineer

React, FastAPI, LangChain, LangGraph, Gemini embeddings, Redis, Docker, AWS ECS

- Admin, Teacher and Student dashboards with an AI Personality Engine that personalizes tutoring to learning style.
- RAG pipeline for context-aware tutoring from uploaded documents; automated quiz generation, lesson planning and remedial learning flows

Agrochemical Demand Forecasting

AI/ML Engineer

Python, LightGBM, XGBoost, Pandas, PostgreSQL, FastAPI, AWS

- End-to-end forecasting on sales, weather, crop and inventory data with engineered time-series features and rolling validation.
- Deployed FastAPI inference APIs with automated retraining and AWS production deployment.

Laboratory Information System

Full Stack Developer

Java, Spring Boot, MySQL, Hibernate, AWS

- Built REST APIs for lab operations, patient handling, and automated reporting; developed Thymeleaf UI improving technician efficiency.
- Implemented Spring Security meeting healthcare-grade compliance; deployed on AWS with automated CI/CD pipelines, ensuring 99.9% system uptime.

EDUCATION

Bachelor of Technology in Computer Science Engineering

RGPV University, Bhopal, Madhya Pradesh, India

2023

CERTIFICATIONS

- ▶ AWS Certified Solutions Architect – Associate
- ▶ AWS Certified Cloud Practitioner